Lesson plan · for the teacher

Asking questions: SELECT, WHERE, ORDER BY

Data engineering · Level 3 · Lesson 2

Time
40 min

Materials
A card per pupil: class, birth month and how they travel (made up or chosen by the pupil); Board and chalk; Optional: a phone or computer for the SQL steps

Learning goals

- Pick columns with SELECT and rows with WHERE.
- Sort with ORDER BY, and keep the top rows with LIMIT.
- Check every answer against what we know.

Activity: Human database

1. Warm-up (5 min): Ask: how would you find every pupil in the school who walks more than 30 minutes, from a sheet of 400 rows? Collect ideas.
2. Teach (8 min): Write a query on the board and label each part: the columns, the table, the condition and the order. Show AND, OR and DESC.
3. Play (15 min): Hand out the cards. Read queries aloud; the matching pupils step forward and line up in order. Include one query that should return nobody.
4. Write (7 min): Pairs write a query on the board; the class runs it and checks that the answer makes sense.
5. Wrap-up (5 min): Ask how the class knew an answer was wrong. Then give the exit ticket.

Let pupils choose or make up their birth month and travel, so nobody has to share real details. Keep the queries short at first, then add AND and ORDER BY.

Discussion

- Which part of a query picks the rows?
- Why did '6a' find nobody?
- How do you get the top three, not the bottom three?

Exit ticket

- Write a query for the pupils in 6B with a science mark over 15, highest first.

Try it at school tomorrow
Play “Human database”: each pupil holds a card with a class, a birth month and how they travel. Read out a query, such as pupils where travel is walking, ordered by birth month. Those who match step forward and line up in order.

Practice questions · for you

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1 **You want the three highest maths marks, biggest first. Which lines do that?**

◯ ORDER BY mark DESC LIMIT 3 ◯ ORDER BY mark LIMIT 3 ◯ WHERE mark = 3

2 **Before the parents' evening: find every pupil in class 6A with a maths mark under 10. Show their name and maths mark.**

```
SELECT name, maths FROM results
WHERE class = '6A';
```

Write your program:

3 **Fill the gaps: the names of the pupils in 6B, best science mark first.**

```
SELECT name FROM results
WHERE class = _____
ORDER BY science _____ ;
```

Word bank: `6B` `ASC` `'6B'` `DESC`

4 **The science prize: show the name and science mark of the three highest science marks, highest first.**

```
SELECT name, science FROM results
ORDER BY science
LIMIT 3;
```

Write your program:

5 **Class 6A has four pupils, but this query returns no rows. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  SELECT name, maths
2  FROM results
3  WHERE class = '6a'
4  ORDER BY maths DESC;
```

Which question about your pupils would you ask a database first? Write it in words, then try it in SQL.

Check your answers

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1 **ORDER BY mark DESC LIMIT 3**

DESC sorts the biggest first, and LIMIT 3 keeps the first three rows.

2

```
SELECT name, maths FROM results
WHERE class = '6A' AND maths < 10;
```

AND keeps only the rows where both conditions are true: Tunde (9) and Kofi (7).

3

```
SELECT name FROM results
WHERE class = '6B'
ORDER BY science DESC;
```

Text in SQL goes in single quotes, '6B'; without them SQLite can't read 6B at all and stops with an error (a word like Grace without quotes would be taken as a column name).

4

```
SELECT name, science FROM results
ORDER BY science DESC
LIMIT 3;
```

ORDER BY on its own sorts from the lowest.

5 **Line 3:** `WHERE class = '6A'`

= on text is exact: '6a' with a small a is not '6A'.